Kubernetes Task-2 - Sudir

Task Description:

Create the K8s EKS,further you have to do the deployment of the Nginx application and access the application outside the cluster.

Techstacks needs to be used :

- **AWS EKS**
- **EKSCtl**
- **Kubectl**

How do I submit my work?

- **Push all your work files to GitHub (Code & O/P screenshot images must).**
- **Submit your URLs in the portal.**

Terms and Conditions?

- **You agree to not share this confidential document with anyone.**
- **You agree to open-source your code (it may even look good on your profile!). Do not mention our company's name anywhere in the code.**
- **We will never use your source code under any circumstances for any commercial purposes; this is just a basic assessment task.**

NOTE: Any violation of Terms and conditions is strictly prohibited. You are bound to adhere to it.

Cluster Creation

Created EKS cluster using eksctl:

eksctl create cluster --name sudir-cluster-1 --region ap-south-1 --node-type t3.medium --nodes 2

```
[root@ip-172-31-44-245 ~]# eksctl create cluster --name sudir-cluster-1 --region ap-south-1 --node-type t3.medium --nodes 2
2026-04-15 11:53:45 [ℹ]  eksctl version 0.225.0
2026-04-15 11:53:45 [ℹ]  using region ap-south-1
2026-04-15 11:53:45 [ℹ]  setting availability zones to [ap-south-1b ap-south-1a ap-south-1c]
2026-04-15 11:53:45 [ℹ]  subnets for ap-south-1b - public:192.168.0.0/19 private:192.168.96.0/19
2026-04-15 11:53:45 [ℹ]  subnets for ap-south-1a - public:192.168.32.0/19 private:192.168.128.0/19
2026-04-15 11:53:45 [ℹ]  subnets for ap-south-1c - public:192.168.64.0/19 private:192.168.160.0/19
2026-04-15 11:53:45 [ℹ]  nodegroup "ng-24b149ff" will use "" [AmazonLinux2023/1.34]
2026-04-15 11:53:45 [!]  Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups
nger be created by default. To maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster confic
age/auto-mode/
2026-04-15 11:53:45 [ℹ]  using Kubernetes version 1.34
2026-04-15 11:53:45 [ℹ]  creating EKS cluster "sudir-cluster-1" in "ap-south-1" region with managed nodes
2026-04-15 11:53:45 [ℹ]  will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2026-04-15 11:53:45 [ℹ]  if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --regic
2026-04-15 11:53:45 [ℹ]  Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluste
2026-04-15 11:53:45 [ℹ]  CloudWatch logging will not be enabled for cluster "sudir-cluster-1" in "ap-south-1"
2026-04-15 11:53:45 [ℹ]  you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HEF
er=sudir-cluster-1'
2026-04-15 11:53:45 [ℹ]  default addons kube-proxy, coredns, metrics-server, vpc-cni were not specified, will install them as E
2026-04-15 11:53:45 [ℹ]
2 sequential tasks: { create cluster control plane "sudir-cluster-1",
    2 sequential sub-tasks: {
        2 sequential sub-tasks: {
            1 task: { create addons },
            wait for control plane to become ready,
        },
        create managed nodegroup "ng-24b149ff",
    }
}
2026-04-15 11:53:45 [ℹ]  building cluster stack "eksctl-sudir-cluster-1-cluster"
2026-04-15 11:53:45 [ℹ]  deploying stack "eksctl-sudir-cluster-1-cluster"
```

Verified cluster:

kubectl get nodes

```
[root@ip-172-31-44-245 ~]# kubectl get nodes
NAME                                          STATUS   ROLES    AGE     VERSION
ip-192-168-25-223.ap-south-1.compute.internal  Ready    <none>   10m     v1.34.4-eks-f69f56f
ip-192-168-41-225.ap-south-1.compute.internal  Ready    <none>   9m58s   v1.34.4-eks-f69f56f
[root@ip-172-31-44-245 ~]#
```

Nginx Deployment:

```
-rw-r--r--. 1 root root      309 Apr 15 12:10 nginx-deployment.yaml
-rw-r--r--. 1 root root      180 Apr 15 12:10 nginx-service.yaml
[root@ip-172-31-44-245 ~]#
```

Created deployment YAML:

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
 name: nginx-deployment
spec:
 replicas: 2
 selector:
  matchLabels:
   app: nginx
 template:
  metadata:
   labels:
    app: nginx
  spec:
   containers:
   - name: nginx
    image: nginx
    ports:
    - containerPort: 80
```

Applied deployment:

kubectl apply -f nginx-deployment.yaml

Verified pods:

kubectl get pods

```
[root@ip-172-31-44-245 ~]# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-7ccccd94f7-f45fw   1/1     Running   0          6m49s
nginx-deployment-7ccccd94f7-fhgwn   1/1     Running   0          6m49s
[root@ip-172-31-44-245 ~]#
```

Exposing Application via Loadbalancer :

Created service YAML:

```
apiVersion: v1
kind: Service
metadata:
 name: nginx-service
spec:
 type: LoadBalancer
 selector:
  app: nginx
 ports:
  - protocol: TCP
   port: 80
   targetPort: 80
```

Applied service:

kubectl apply -f nginx-service.yaml

Accessing Application

Retrieved external DNS:

kubectl get svc nginx-service

```
NAME            TYPE           CLUSTER-IP      EXTERNAL-IP                                                                PORT(S)        AGE
kubernetes      ClusterIP      10.100.0.1      <none>                                                                     443/TCP        18m
nginx-service   LoadBalancer   10.100.46.227   a71edf62b329749b680926bae300d0a0-1617189802.ap-south-1.elb.amazonaws.com   80:30381/TCP   6m38s
[root@ip-172-31-44-245 ~]#
```

Accessed application via browser:

http://a71edf62b329749b680926bae300d0a0-1617189802.ap-south-1.elb.amazonaws.com/